Truss Grid Analyzer

Quick Start

Your first rig in ten minutes

Written for version 1.28.0
12 October 2026

By **G.E. Simmons Falk**. Truss Grid Analyzer is an expansion, for indeterminate grids, of *Truss Load Analyzer - EOT* by **Delbert L. Hall and Jon Sogoian**, the originators of the program.

For everything else, see the full User Guide (PDF, or the searchable HTML copy). The guides are written for version 1.28.0 and are not updated for every release.

1 Before You Start

Truss Grid Analyzer works out the load on every chain hoist and checks every truss in an entertainment truss rig, from a single truss on two hoists up to a grid of trusses bolted together at corner blocks. You draw the trusses, put the loads on them, place the hoists, and it shows you, as you type, whether each hoist and each truss is within the manufacturer's limits.

> **A calculation aid, not an engineering approval**
> Check the results against the manufacturer's current data, and have a qualified person review every rig before it flies. You are responsible for checking the results before you use them to make rigging decisions.

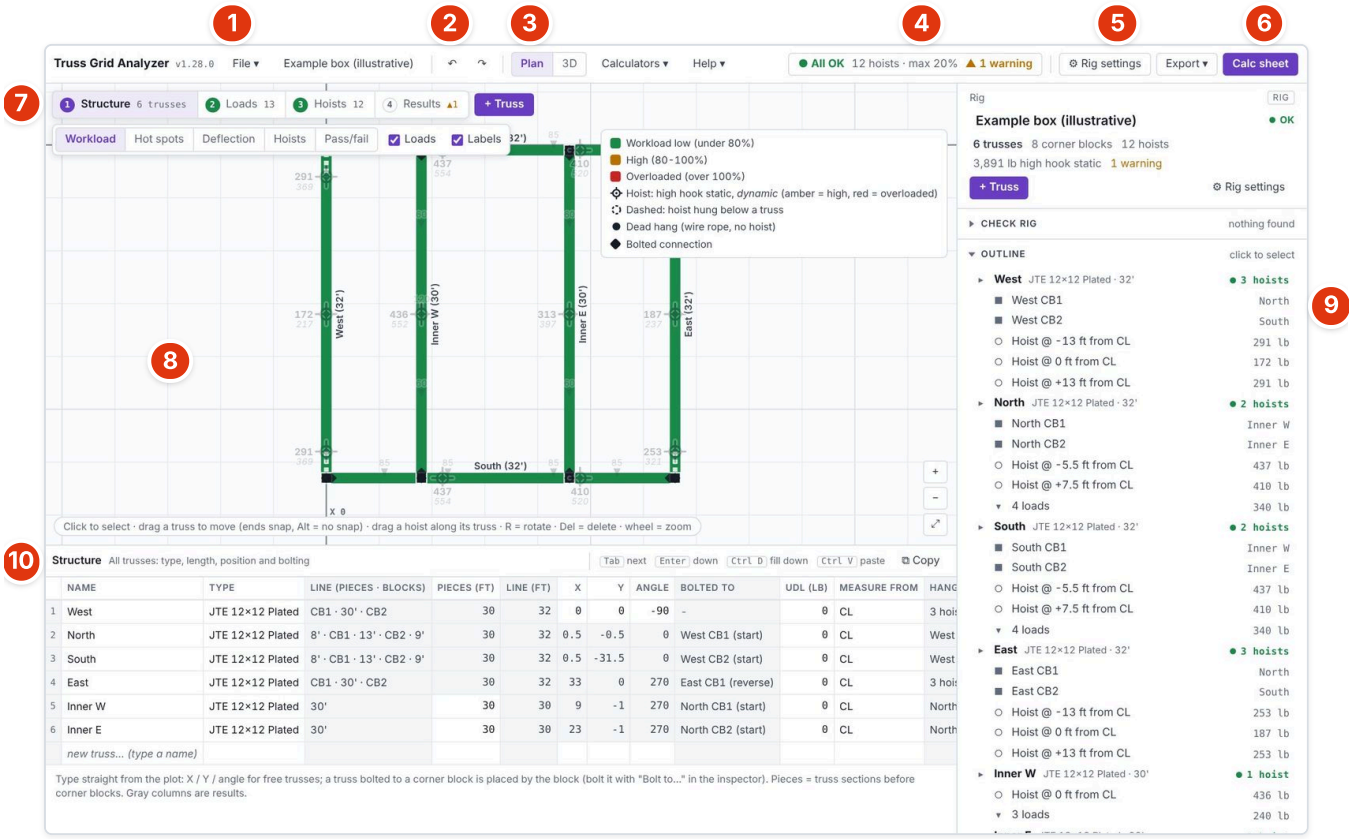
> **Indeterminate rigs (ANSI E1.6-3)**
> ANSI E1.6-3 (*Selection and Use of Serially Manufactured Chain Hoists in the Entertainment Industry*) calls a rig **indeterminate** when it hangs from more than two hoists in a straight line, or more than three hoists not in a line - a truss on three hoists, or almost any grid or box. On an indeterminate rig the load on each hoist depends on how stiff the trusses, the rigging and the structure above are, and load can shift between hoists as they run. The standard (section 7.1) accepts a load calculation as a way to reduce the risk of overload only when every weight is known and accounted for, the calculation method is accurate, and any load shifting will not be enough to cause an overload. This app's whole-rig analysis is built for these rigs, but it is an estimate: allow for load shifting, level the hoists carefully, and verify loads (for example with load cells) where the margin is small.

1.1 Open the App

1. The whole app is one file, `index.html` (on GitHub it is `dist/index.html`). Save it anywhere on your computer.
2. Double-click it. It opens in your web browser (Chrome, Edge, Firefox or Safari). No install, no account, no internet connection needed, and nothing is sent anywhere.
3. The first time, it shows an **example box** so you can look around. The version number is next to the name, top left (this guide: **v1.28.0**).

> **Your work is kept in this browser only**
> Every change is saved automatically inside the browser on this computer. Clearing the browser's data, or opening the file in a different browser, gives you a different (or empty) workspace. Use **File > Save** to keep a rig as a `.rig.json` file you can back up, email and open again.

1.2 The Screen



The app with the example box loaded (light theme; the app starts in the dark theme - change it under **File > Theme**).

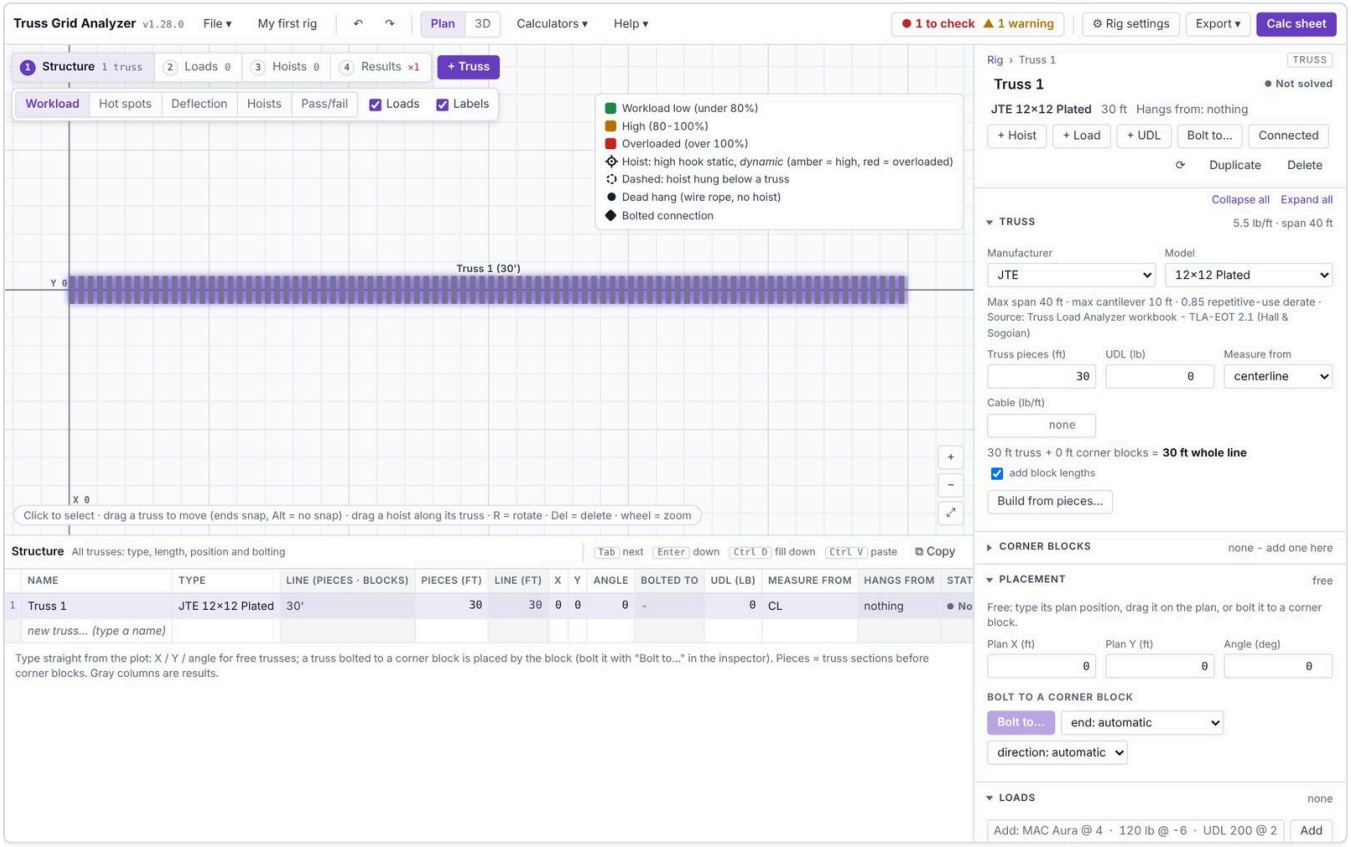
1. **File** - new rig, open, save, the example box, theme, About.
2. **Undo / Redo** - `Ctrl` + `Z`, `Ctrl` + `Y`.
3. **Plan / 3D** - switch between the plan and a 3D view.
4. **Status** - "All OK" or how many things to check, the busiest hoist, warnings. Click it for the results.
5. **Rig settings** - dynamic factor, units (feet or meters), allowances and more.
6. **Calc sheet** - a printable calculation, for independent review of the software's work.
7. **The four steps** - 1 Structure, 2 Loads, 3 Hoists, 4 Results. Use them in any order, any time.
8. **The plan** - the rig to scale. Click anything to select it; drag trusses and hoists.
9. **Side panel** - everything about what you clicked: the rig, a truss, a hoist, a load or a corner block.
10. **The grid** - a spreadsheet of the current step: every truss, every load or every hoist.

2 Your First Rig

This walk-through builds a **30 ft truss hung on two 1-ton hoists, with six moving lights**. It takes about ten minutes. Every number below is what the app shows for version 1.28.0.

2.1 Step 1 - Structure: Add the Truss

1. Choose **File > New rig** and click **OK**. (A new rig replaces the one on screen. Undo brings it back until you close or reload the page, but save anything you want to keep first.)
2. Click the rig name at the top (*Untitled rig*) and type **My first rig**.
3. Click **+ Truss**. A 20 ft truss called *Truss 1* appears on the plan and in the side panel.
4. In the side panel, under **Truss**, pick the **Manufacturer** and **Model** you use. This example keeps the default, **JTE 12×12 Plated**.
5. In **Truss pieces (ft)**, type **30** and press `Enter`. The truss is now 30 ft long.



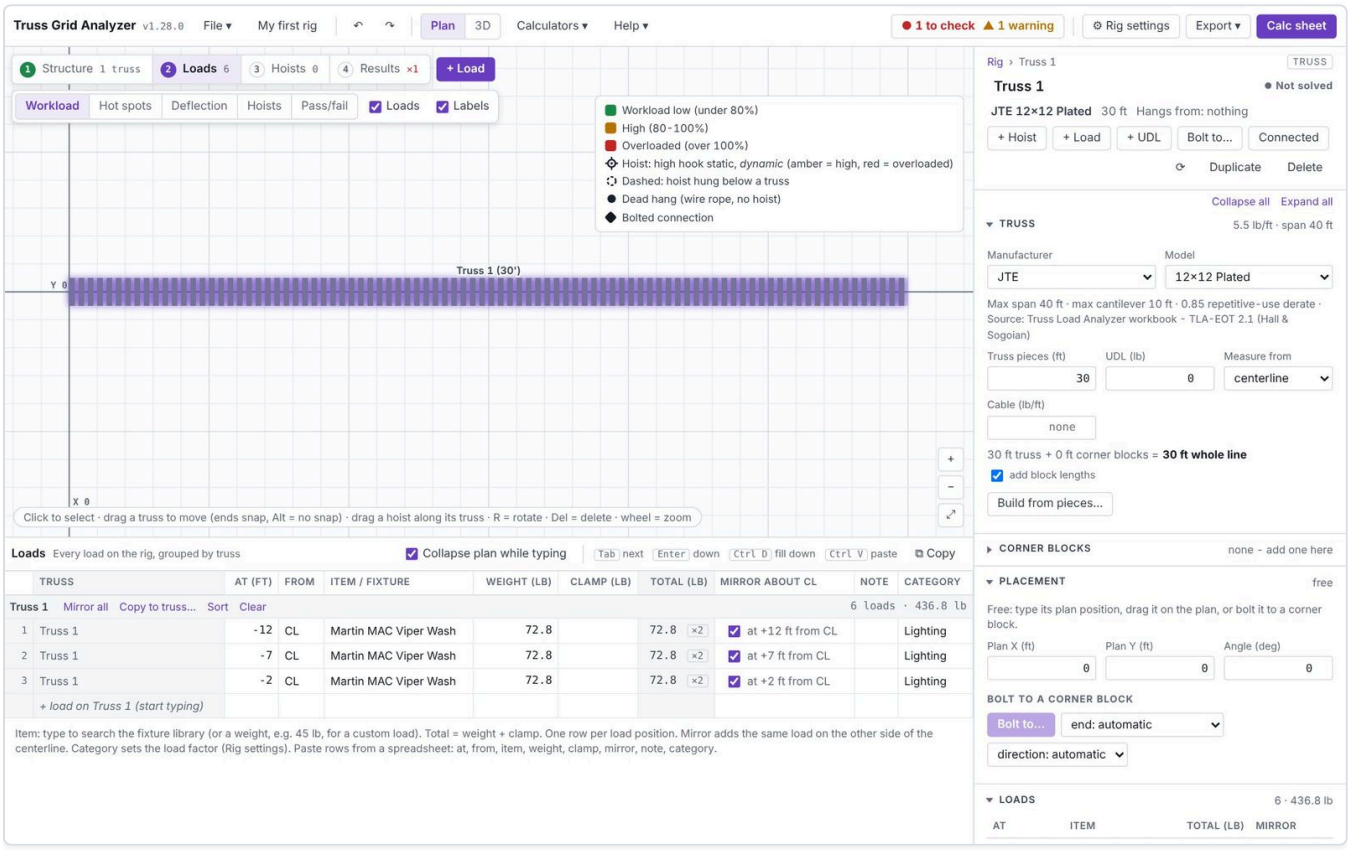
Step 1. One 30 ft truss. It shows *Not solved* and the status says *1 to check* because it has no hoists yet - that is expected.

> **Typing lengths**
> Type decimal feet (`12.5`) or feet and inches (`12-6` , `12'6"` , `12' 6 1/2"` , `12 ft 6 in`). In metric (**Rig settings > Units**) type meters (`3.8`), `380 cm` or `3800 mm` .

2.2 Step 2 - Loads: Hang Six Lights

A new truss measures positions from its **centerline** (CL): `-12` is 12 ft before the middle, `+12` is 12 ft after it.

1. Click step **2 Loads** (or press `2`).
2. Click the truss on the plan. In the side panel, open **Loads**. In the box that says *Add: fixture or weight*, type **MAC Viper Wash @ -12** and press `Enter`. The app finds the fixture in its library (72.8 lb) and places it 12 ft left of the centerline.
3. Do the same with **MAC Viper Wash @ -7** and **MAC Viper Wash @ -2**.
4. Click **Mirror all**. Each light gets a twin on the other side of the centerline (at +12, +7 and +2), so there are now six.



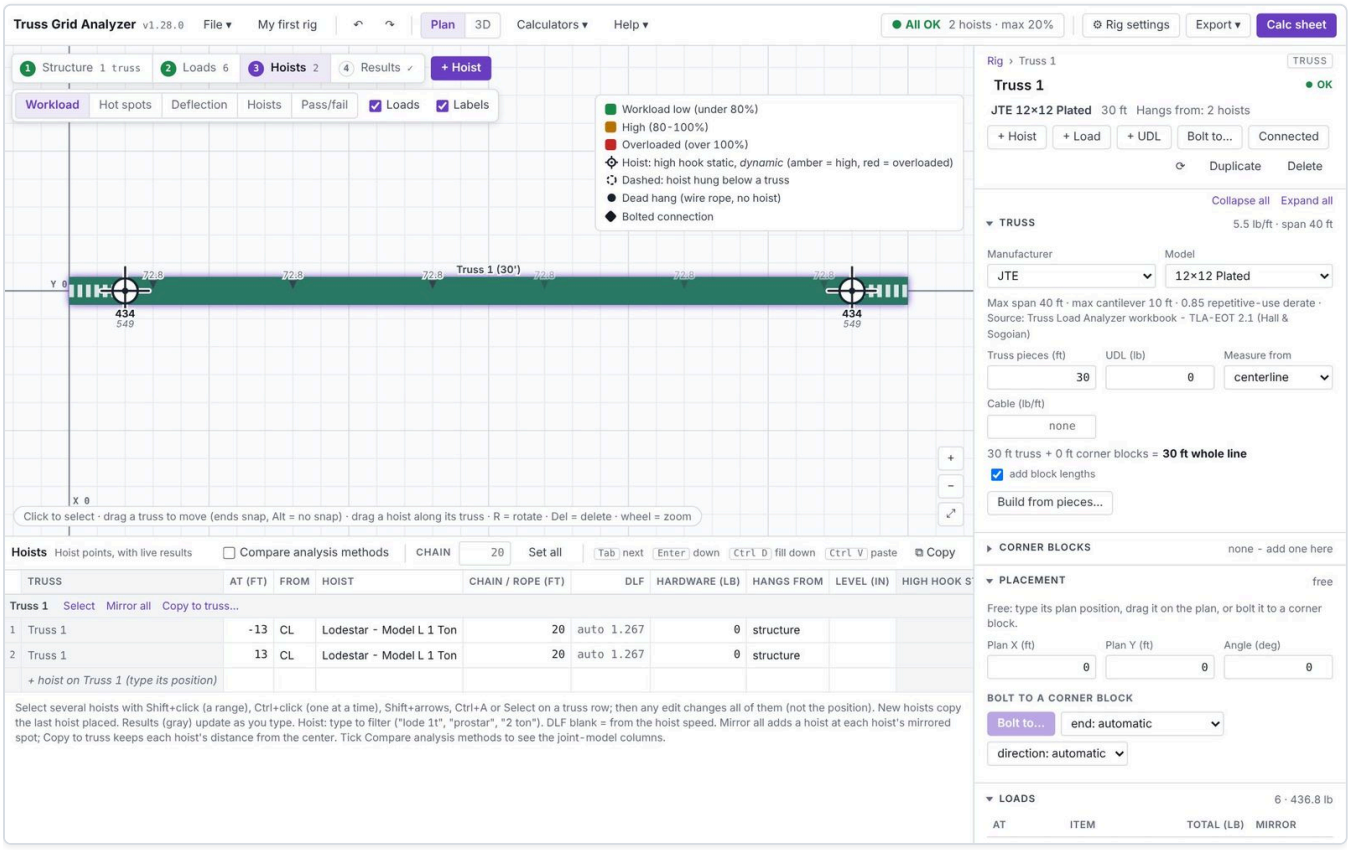
Step 2. Three loads, each mirrored (×2 in the grid). The grid under the plan lists every load; you can type there too.

For a load that is not in the library, type its weight instead: **120 lb @ 6**. You can also type straight into the **Loads** grid: the *Item / fixture* cell searches the library as you type.

2.3 Step 3 - Hoists: Hang It Up

1. Click step **3 Hoists** (or press `3`).
2. In the grid under the plan, find the row *+ hoist on Truss 1 (type its position)* and click its **At (ft)** cell. Type **-13** and press `Enter`. A hoist appears 13 ft left of the centerline (2 ft in from the end).

3. The cursor is now back on the *+ hoist* row, so just type **13** and press `Enter` for the second hoist.
4. The **Hoist** column shows the model (the first one is a 1-ton CM Lodestar with 20 ft of chain). To change it, click the cell and type part of a name - `lode 1t`, `prostar`, `2 ton` - then pick from the list. Set the chain length in the **Chain** column.

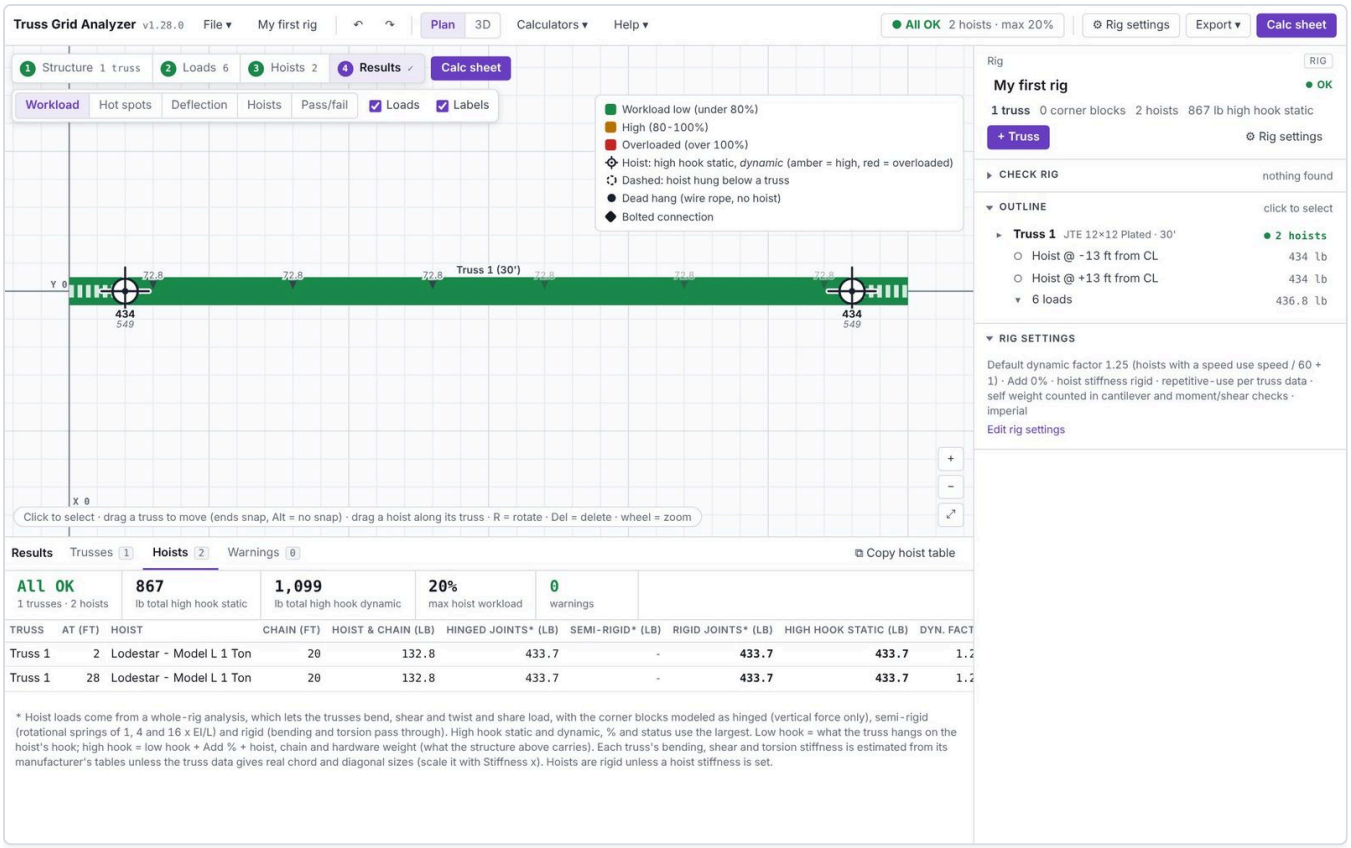


Step 3. Two hoists. On the plan each one shows its high hook static load (434) and, in italics, its dynamic load (549).

> **Other ways to add hoists**
>
> Select the truss, click **+ Hoist** above the plan (it adds one in the middle of the selected truss - with no truss selected it adds a new truss instead), then drag it along the truss on the plan - it moves in 1-inch steps and shows its position as you drag.

2.4 Step 4 - Results: Read Them

Click step **4 Results** (or press `4`, or click the status at the top).



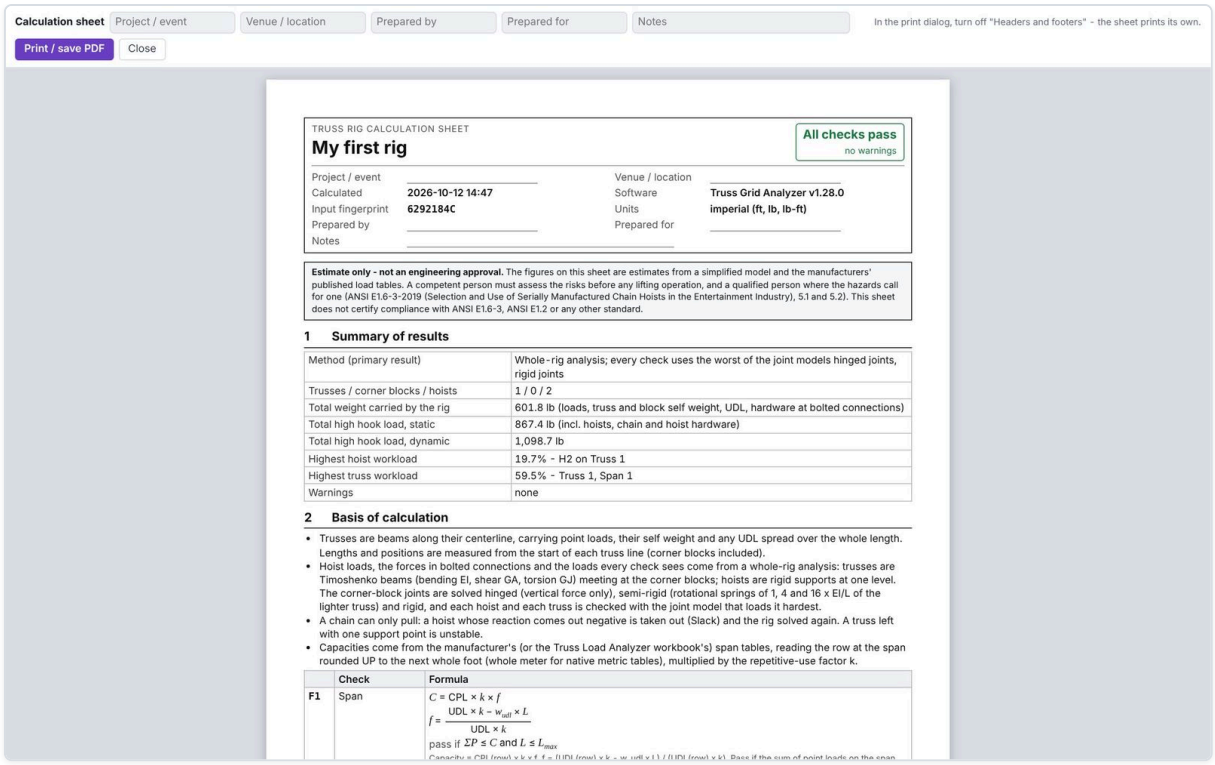
Step 4. All OK. Each hoist carries 433.7 lb static and 549.4 lb dynamic, 20% of its 2,200 lb capacity.

What the numbers mean for this rig:

Figure	Value	What it is
Low hook load	300.9 lb	What the truss hangs on each hook: half of 6 × 72.8 lb lights + 30 ft × 5.5 lb/ft of truss.
Hoist & chain	132.8 lb	The hoist body plus 20 ft of chain.
High hook static	433.7 lb	Low hook + hoist and chain: what the structure above carries. This is checked against the hoist's capacity.
High hook dynamic	549.4 lb	Static × the dynamic factor (16 fpm hoist: 16 / 60 + 1 = 1.267).
Workload	20%	High hook static ÷ capacity (2,200 lb).

2.5 Save It and Print the Calculation

1. **File > Save** downloads `My first rig.rig.json`. Keep it - **File > Open...** brings it back on any computer.
2. Click **Calc sheet** (top right). Fill in the project, venue, prepared by and prepared for boxes at the top, then click **Print / save PDF** and choose *Save as PDF* (or a printer). Press `Esc` or **Close** to return.



The calculation sheet: every input, formula, table look-up and check, set out so the software's work can be reviewed independently.

3 Reading the Results

3.1 Colors and Statuses

Color / status	Meaning	What to do
OK	Under 80% of what the manufacturer allows.	Nothing.
High	80-100%. Within limits but little margin.	Consider it; check your weights are right.
Overloaded	Over 100% of a limit (a hoist's capacity, a span or cantilever table, bending moment, shear, or the maker's deflection limit).	Must be fixed: move or lighten loads, add or move hoists, or use a stronger truss or hoist.
Too long	A span is longer than the truss table allows, or a cantilever is longer than a quarter of the maximum span.	Add a support or shorten the overhang.
Slack	The load would push a hoist *up*. A chain can only pull, so it carries nothing and the rig is solved without it.	Move the hoist or the loads; the other hoists now carry more.
Unstable	A truss or assembly is not held up properly (for example one hoist only, or all hoists in a line) and would tip.	Add hoists. Do not use its numbers.
Not solved	The truss has no supports yet, or depends on something missing.	Add hoists or bolt it to another truss.

3.2 Where to Look

- **The status at the top** sums up the whole rig. Click it to jump to the results.
- **Step 4 Results** has three tabs: **Trusses** (every check for every truss), **Hoists** (every hoist's loads), and **Warnings** (everything the app wants you to look at). Click a row to select that truss or hoist.
- **Click a truss** and open **Checks** in the side panel: a drawing of the truss with its supports and loads, where it works hardest, its reactions, shear and moment diagrams, deflection, and a bar for each check.
- **Click a hoist** to see its low hook, high hook static and high hook dynamic loads, and a table of where its load comes from.
- **Check rig** (click an empty part of the plan to show the rig panel) lists likely input mistakes: a load with no weight, two hoists at one point, bolted ends that don't meet, too few hoists.

> **Level sensitivity**
> On short, stiff spans a hoist running just 1/4" high or low can change its load by hundreds of pounds. The app warns when that is more than 10% of a hoist's capacity - the example box shows this warning. Level those hoists carefully.

4 Handy to Know

4.1 Keyboard

Key	Does	Key	Does
`1` `2` `3` `4`	Go to a step	`F`	Fit the rig in view
`Ctrl`+`Z` / `Ctrl`+`Y`	Undo / redo	`R`	Rotate the selected truss 90°
`Delete`	Delete the selected load, hoist or truss	`Esc`	Cancel / close
`Tab` / `Enter`	In a grid: next cell / next row	`Ctrl`+`D`	In a grid: copy the cell above
`Ctrl`+`V`	In a grid: paste rows from Excel	Mouse wheel	Zoom the plan; drag empty space to pan

4.2 Building a Grid or a Box

Trusses join at **corner blocks**. A corner block belongs to one truss (add it in that truss's **Corner blocks** section: **+ Corner block at start**, **+ Corner block at end**, or at a distance). Then select another truss, click **Bolt to...** and click the corner block on the plan. The bolted truss moves to the block and follows it from then on. **File > Load example box** shows a complete box built this way. See the User Guide, chapter *Step 1 - Structure*, for the details.

4.3 Menus You Will Use

- **Calculators** - circular truss, simple UDL, fixture weights, and the databases of trusses, corner blocks and hoists (where you can add your own).
- **Export** - copy the hoist table for Excel, save the view as a picture, print the screen.
- **⚙ Rig settings** - units, dynamic factor, *Add %*, cable allowance, load factors, deflection limit, calc sheet paper size.
- **Help** - this Quick Start and the full User Guide.
- **File > About, method & version history** - how the app calculates, the license, and what changed in each version.

4.4 Credits

Truss Grid Analyzer is by G.E. Simmons Falk. It is an expansion of *Truss Load Analyzer - EOT* by Delbert L. Hall and Jon Sogoian, who are the originators of the program, for indeterminate grids. It is free and must never be sold.